

Forensics Principles & Imaging

The Golden Rules

Preserve the evidence – work on COPIES, never the original
Document everything – chain of custody, timestamps, commands
Least invasive first – collect volatile data before it dies
Hash everything – SHA-256 the image; prove it didn't change

```
sha256sum evidence.img      # record; recheck later
dd if=/dev/sda of=evidence.img bs=4M status=progress # image a disk
```

Order of Volatility (Collect FAST Things First)

Memory dies fastest – collect in this order:

1. RAM (memory dump) <-- THE most perishable
2. Network connections/state <-- connection table
3. Running processes
4. Temporary files / cache
5. Disk (persistent)

Why it matters: pulling the plug loses RAM forever. The whole attacker session (processes, passwords, crypto keys in memory) is sitting there – until you reboot.

Write-Blocking

You must never write to evidence storage while imaging:

Hardware write-blocker, OR
Boot examiner system, image source read-only, OR

`dd` with `conv=noerror, sync` reading from a source you've mounted read-only

Without write-blocking, the "evidence" can be challenged: "your own tools modified the disk."

Imaging Tools That Matter

`dd` / `ddrescue` – raw images (the gold standard)

`guymager`, `FTK Imager`, `Autopsy` – GUI-friendly

`Magnet RAM Capture`, `dumpit` (Win), `lime` (Linux) – memory capture

The Analyst's Mindset

Forensics = "tell the story with proof":

Evidence without timestamps = weak evidence

No chain of custody = evidence that can't be presented

No hashes = can't prove integrity

DFIR in 2026: attackers LIVE in memory (fileless attacks, LOLBins - Living Off the Land Binaries). **Memory forensics is now the default, not the bonus.** Masters of it run circles around the disk-only analysts.

[← Module start](#)

[Memory Forensics with Volatili...](#)